

Experimental Setup

Variables

Independent Variables

- The cognitive load level of secondary activity (besides the primary task)

Dependent Variables

- Accuracy
 - Veracity of a headline → True/False
- Confidence
 - Confidence while evaluating a statement → 1 – 7 Likert Scale
 - Confidence in direct forwarding/sharing → 1 – 7 Likert Scale

Confounding Variables

- AI Literacy → accuracy:
 - Personas with higher AI literacy may be more aware that AI-generated content can be fluent but false, making them more careful when judging AI-generated misinformation.
- Education Level → accuracy & confidence:
 - Education may influence reading comprehension, critical thinking, and familiarity with scientific, political, economic, or technological claims.
- General confidence → confidence:
 - A generally confident or overconfident persona may report high certainty without any basis, even when its true/false judgment is incorrect.
- Special Knowledge → accuracy & confidence:
 - Personas with stronger knowledge in a domain may be better able to identify whether a statement in that domain is credible.
- Trust in AI (scepticism and naivety) → accuracy & confidence:
 - High trust in AI may reduce skepticism toward AI-generated statements and increase willingness to accept or share them.
- Resilience (the ability to maintain performance under distraction or cognitive strain) → accuracy & confidence:
 - More resilient personas may be less affected by the secondary cognitive load tasks.
- Cognitive capacity (the persona's reasoning ability, working memory, or ability to think analytically) → accuracy & confidence:
 - Higher cognitive capacity may help the persona complete secondary tasks while still accurately evaluating the target statement.

Demographic Variables Collected for Reproducibility

- Age

- Gender
- Country

Experimental Design

1. Pre-survey:
 - a. AI-Literacy Test
 - b. General Confidence Test
2. Experimental phase
 - a. Use a between-subject design:

Group	Condition	Task (see "Cognitive Load Tasks")
Group 0	No load	T7 only
Group 1	Low load	T1 + T7
Group 2	High load	T1 + T2 + T3 + T7

- b. Assess 10 pieces of AI (mis)-information from a variety of domains (politics + environment + health + economics + technology) x (true + false) → use 10 prompts for each persona
- c. Respond to the veracity and confidence questions using JSON format

Response Fields

1. Veracity

Q: Do you think the statement is true or false?

Options: True/False

Q: Briefly explain the main reason for your judgment in one sentence.

Options: <text input field>

2. Confidence

Q: How confident are you in your true/false judgment?

Options:

1 = not at all confident

2 = slightly confident

3 = somewhat confident

4 = moderately confident

5 = quite confident

6 = very confident

7 = extremely confident

Q: If you saw this statement on social media, how likely would you be to share it?

Options:

1 = extremely unlikely to share

2 = unlikely to share

3 = somewhat unlikely to share

4 = neither unlikely nor likely

5 = somewhat likely to share

6 = likely to share

7 = extremely likely to share

Cognitive Load Task

T1 = remove tags from neutral sentence

T2 = reverse reconstructed neutral sentence

T3 = write numbers from negative five to positive five in words

T7 = evaluate target statement

Condition	Description	Prompt structure
No external cognitive load	Persona directly evaluates the target statement	Statement evaluation only
Low external cognitive load	Persona completes one irrelevant secondary task before evaluation	Tag removal + statement evaluation
High external cognitive load	Persona completes three irrelevant secondary tasks before evaluation	Tag removal + reverse word order + number sequence + statement evaluation

(Mis)information dataset

- Intel Misinformation Guard Dataset
- 100 pieces of information: 50 true + 50 false
- 5 different domains generalization
- Example:

Domain	True	False
Health/Medicine	"A diet rich in fruits, vegetables, and whole grains can help reduce the risk of chronic diseases, such as heart disease and diabetes."	"Eating a strict diet of only pineapple and bananas for 30 days will reset your metabolism and cure chronic diseases."
Politics/Governme	"The 9/11 attacks were carried out by a	"Eating a strict diet of only pineapple and

nt/Conspiracy	combination of Al-Qaeda operatives and the US government, with the latter providing support and intelligence to the former."	bananas for 30 days will reset your metabolism and cure chronic diseases."
Finance/Economics	"Investing in a diversified portfolio with a mix of low-risk and high-risk assets can help mitigate potential losses and achieve long-term financial goals."	"Buying gold coins is a foolproof way to protect your savings from market crashes."
Science/Environment/Climate	"Solar panels can generate electricity from sunlight without emitting greenhouse gases or pollutants."	"The Earth's temperature has been steadily decreasing over the past century due to a natural cooling trend, and human activities have no significant impact on climate change."
Technology/AI/Digital Platforms	"5G networks use a higher frequency range than previous cellular technologies, which allows for faster data transfer rates and lower latency."	"AI systems have achieved human-like sentience and are capable of self-awareness."

Prompts

a) Group 0:

You are an AI agent simulating a young adult participant. Respond according to the persona profile below.

Persona profile:
{PERSONA_PROFILE}

You will evaluate one information statement. Do not search the internet. Use only your own judgment.

Statement ID: {STATEMENT_ID}

Statement:
{TARGET_STATEMENT}

Answer the following questions:

1. Do you think the statement is true or false?
2. How confident are you in your true/false judgment? Use a 1 to 7 scale, where 1 means not at all confident and 7 means extremely confident.
3. If you saw this statement on social media, how likely would you be to share it? Use a 1 to 7 scale, where 1 means extremely unlikely to share, and 7 means extremely likely to share.
4. Briefly explain the main reason for your true/false judgment in one sentence.

Return valid JSON only:

```
{
  "statement_id": "{STATEMENT_ID}",
  "condition": "CLO_no_load",
```

```
"veracity": "True/False",  
"evaluation_confidence": 1-7,  
"sharing_intention": 1-7,  
"brief_reason_for_veracity": "one sentence"  
}
```

b) Group 1:

You are an AI agent simulating a young adult participant. Respond according to the persona profile below.

Persona profile:
{PERSONA_PROFILE}

You will first complete one short secondary task. After that, you will evaluate one information statement. Do not search the internet. Use only your own judgment.

Secondary task 1:
Remove the [SEG] and [/SEG] tags from the following sentence.

Tagged sentence:
[SEG]The[/SEG] [SEG]orange[/SEG] [SEG]notebook[/SEG] [SEG]was[/SEG]
[SEG]placed[/SEG] [SEG]beside[/SEG] [SEG]the[/SEG] [SEG]window[/SEG]
[SEG]yesterday[/SEG].

Now evaluate the statement below.

Statement ID: {STATEMENT_ID}

Statement:
{TARGET_STATEMENT}

Answer the following questions:

1. Do you think the statement is true or false?
2. How confident are you in your true/false judgment? Use a 1 to 7 scale, where 1 means not at all confident and 7 means extremely confident.
3. If you saw this statement on social media, how likely would you be to share it? Use a 1 to 7 scale, where 1 means extremely unlikely to share and 7 means extremely likely to share.
4. Briefly explain the main reason for your true/false judgment in one sentence.

Return valid JSON only:

```
{  
  "statement_id": "{STATEMENT_ID}",  
  "condition": "CL1_low_load",  
}
```

```
"secondary_task_1": "removed tags sentence here",  
"veracity": "True/False",  
"evaluation_confidence": 1-7,  
"sharing_intention": 1-7,  
"brief_reason_for_veracity": "one sentence"  
}
```

c) Group 2:

You are an AI agent simulating a young adult participant. Respond according to the persona profile below.

Persona profile:
{PERSONA_PROFILE}

You will first complete three secondary tasks. After that, you will evaluate one information statement. Do not search the internet. Use only your own judgment.

Secondary task 1:
Remove the [SEG] and [/SEG] tags from the following sentence.

Tagged sentence:
[SEG]The[/SEG] [SEG]orange[/SEG] [SEG]notebook[/SEG] [SEG]was[/SEG]
[SEG]placed[/SEG] [SEG]beside[/SEG] [SEG]the[/SEG] [SEG]window[/SEG]
[SEG]yesterday[/SEG].

Secondary task 2:
Write the reconstructed sentence from secondary task 1 in reverse word order.

Secondary task 3:
Write the numbers from negative five to positive five in words.

Now evaluate the statement below.

Statement ID: {STATEMENT_ID}

Statement:
{TARGET_STATEMENT}

Answer the following questions:

1. Do you think the statement is true or false?
2. How confident are you in your true/false judgment? Use a 1 to 7 scale, where 1 means not at all confident and 7 means extremely confident.
3. If you saw this statement on social media, how likely would you be to share it? Use a 1 to 7 scale, where 1 means extremely unlikely to share and 7 means extremely likely to share.

4. Briefly explain the main reason for your true/false judgment in one sentence.

Return valid JSON only:

```
{
  "statement_id": "{STATEMENT_ID}",
  "condition": "CL2_high_load",
  "secondary_task_1": "removed tags sentence here",
  "secondary_task_2": "reconstructed sentence in reverse word order here",
  "secondary_task_3": "numbers in words here",
  "veracity": "True/False",
  "evaluation_confidence": 1-7,
  "sharing_intention": 1-7,
  "brief_reason_for_veracity": "one sentence"
}
```

Analysis

- Primary G*Power analysis:
 - F tests → ANOVA: Fixed effects, omnibus, one way
 - Groups: 3
 - Effect size f: 0.25
 - α : 0.05
 - Power: 0.80
 - Required total sample size: about 158 personas
 - Required per group: about 53 personas
- Accuracy:
 - A response is scored as 1 if the persona's True/False judgment matches the ground truth label of the statement, and 0 otherwise.
- Mean accuracy per persona:
 - Because each persona evaluates 10 statements, mean accuracy is calculated as the number of correct judgments divided by 10.
- Evaluation confidence:
 - The confidence score is recorded as an integer from 1 to 7.
- Sharing intention:
 - The sharing intention score is recorded as an integer from 1 to 7.
- Brief reason:
 - The brief reason is not treated as a main dependent variable. It is used for validation and qualitative inspection.